DRUGS · ANTIBIOTICS: OTHER CLASSES

Azithromycin

Azithromycin is a macrolide with a very long tissue half-life, so short courses of 3 days give a week or more of tissue levels. It is an alternative to clarithromycin in penicillin allergy and an alternative long-term anti-inflammatory macrolide in chronic rhinosinusitis and bronchiectasis. It interacts less through CYP3A4 than clarithromycin but still prolongs the QT interval.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

AZITHROMYCIN

	SUMMARY
Class	Macrolide (azalide)
How it works	50S ribosomal inhibition; immunomodulatory at low long-term doses
Spectrum	As clarithromycin, with better activity against H. influenzae and Moraxella and less against streptococci and staphylococci; atypicals; Chlamydia trachomatis
ENT uses	Penicillin allergy: respiratory infection; long-term anti-inflammatory therapy in chronic rhinosinusitis and in patients with bronchiectasis or cystic fibrosis; pertussis
Adult dose	500 mg once daily for 3 days; long-term 250 mg once daily or 500 mg three times a week
Child dose	10 mg/kg once daily for 3 days (maximum 500 mg)
Avoid	QT prolongation; hepatic impairment
Pregnancy	Use only if no alternative

2 CHAPTER TWO How it works

Mechanism and handling.

2.1 Mechanism

The 15-membered azalide ring improves acid stability and Gram-negative penetration. Its antibacterial and anti-inflammatory actions are those of other macrolides.

2.2 Handling

Takes up into tissues and phagocytes, from which it is slowly released; terminal half-life is about 68 hours. Mostly excreted unchanged in bile. Antacids reduce absorption.

3 CHAPTER THREE Using it in ENT

Indications and doses.

3.1 Acute infection

A 3-day course is equivalent to longer courses of other agents for acute otitis media and sinusitis in penicillin allergy, but pneumococcal and streptococcal macrolide resistance is a real cause of failure.

3.2 Long-term use

For chronic rhinosinusitis, especially with lower-airway disease, azithromycin three times a week is an option when clarithromycin interacts with other drugs. Long-term use selects macrolide resistance and has been linked to **hearing loss** (usually reversible), so check hearing if symptoms occur.

4 CHAPTER FOUR Safety

What goes wrong.

4.1 Adverse effects

- Gastrointestinal upset; hepatitis; hearing loss and tinnitus with long-term use.
- QT prolongation and increased risk of cardiovascular death in patients with cardiac disease.
- Rarely, hypersensitivity including DRESS.

4.2 Interactions

Weaker CYP3A4 effect than clarithromycin, but still caution with warfarin, ciclosporin, digoxin and other QT-prolonging drugs; antacids reduce absorption.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- Why a 3-day course works (tissue half-life).
- Differences from clarithromycin in spectrum and interactions.
- Long-term macrolide risks: QT, resistance, hearing.

5.2 Common pitfalls

COMMON PITFALLS

- Using azithromycin for streptococcal tonsillitis when penicillin can be given.

- Long-term therapy without an ECG baseline in patients at cardiac risk.

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